
1. DIY TOKAMAK

1.1 Background: Building a Shield

A tokamak is a machine that confines a plasma using magnetic fields in a donut shape that scientists call a torus. In a tokamak, magnetic field coils confine plasma particles to allow the plasma to achieve the conditions necessary for fusion.

Our goal in this activity is to decide what material might work best for the shield of the tokamak. The shield keeps the walls and magnets safe from the extremely high temperature plasma used to achieve fusion. It also keeps the operating personnel outside of the vessel safe from high energy neutron damage from the fusion plasma!

Because the shield protects the delicate equipment (and people!), the efficiency of the neutron shield defines the maximum operational lifetime of the toroidal field magnets, and how safe the personnel is.

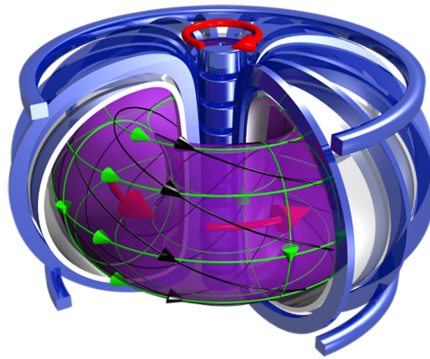


Figure 1: Tokamak Shape

1.2 The First Challenge: Heat Flux

The heat-flux challenge results from the extreme energy density within the fusion plasma:

Up to 150 million degrees Celsius! The first-wall surface materials will experience steady-state heat-fluxes of 10-20 MW/m²- ten times the heat load of a spacecraft re-entering Earth's atmosphere!

Behind the first wall are coolant pipes that will move the heat away from walls. We need materials that will easily transfer heat from the wall to coolant lines within the shield. In other words, materials with a high thermal conductivity.

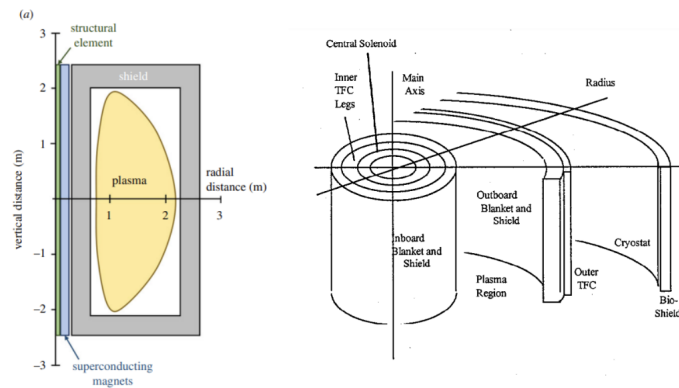


Figure 2: Shield Designs

1. Look up the thermal conductivities of different materials. Is there a material you think would be best for the heat flux challenge?
2. What might be a drawback of the material you chose? Is it expensive? Hard to find?

1.3 The Second Challenge: Irradiation

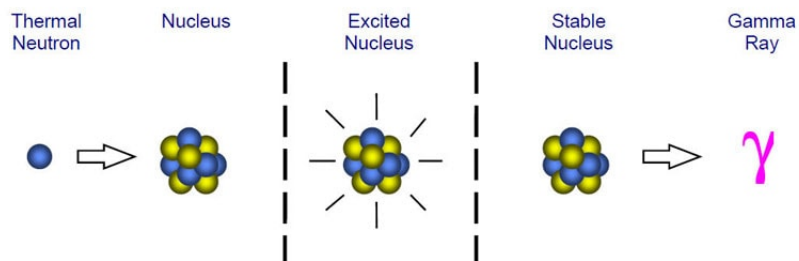


Figure 3: Neutron Activation

The second challenge of irradiation results from 14.1 MeV neutrons released in the deuterium–tritium reaction happening in the reactor. These high energy neutrons can lead cause neutron activation. This damage can cool the plasma, release gamma rays, and erode the shield.

Neutron activation is the process in which neutron radiation induces radioactivity in materials, and occurs when atomic nuclei capture free neutrons, becoming heavier and entering excited states. The excited nucleus decays immediately by emitting gamma rays, or other particles. These products degrade a materials properties, so it is important that the radioactivity of these products is not long-lived! Thus, good candidate shield materials should be restricted to low-activation elements.

3. Look up the low-activation materials (alloys with short lived decay products). Is there a material you think would be best for the irradiation challenge?

4. What might be a drawback of the material you chose? Is it expensive? Hard to find? Is it different than the first material you chose?

Irradiation can also cause defects in materials, which erode the shield and makes it less effective. These defects happen if atoms are displaced. The movement of dislocations allow atoms to slide over each other at low stress levels and is known slip. Shield designs must therefore employ inherently radiation tolerant crystal structures, engineered to provide large numbers of defect trapping sites.

These inclusions are called “dispersoids” and are small and numerous finely divided particles of one substance dispersed in another, like carbon in steel, which prevent the movement of defects.

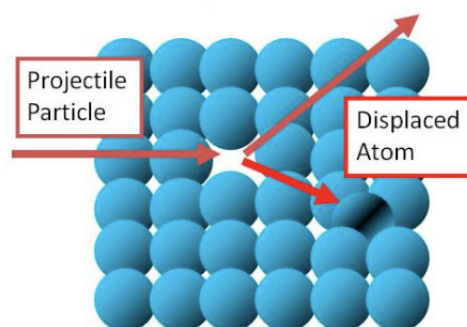


Figure 4: Defect Formation

About steel: Iron is the base metal of steel. In pure iron, the crystal structure has relatively little resistance to the iron atoms slipping past one another, and so pure iron is quite ductile, or soft and easily formed. In steel, small amounts of carbon within the iron act as hardening agents that prevent the movement of defects.

Atomic Structure of Steel

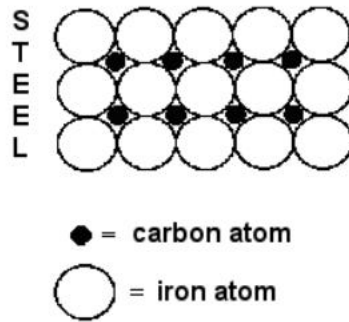


Figure 5: Steel Structure

5. Steel seems like a good choice- but it can corrode. We can add other elements to make steel more resistant: look up an element we can add. In other words, write down the "The Best Alloying Elements for Stronger Steel"

1.4 The Third Challenge: Sputtering

When atoms are ejected, or sputtered, from the shield wall metal and enter the plasma they are stripped of their electrons and cause radiative cooling of the plasma. (Radiation is produced by the deceleration of a charged particle deflected by another charged particle, causing cooling)

We really don't want this! Fusion require super hot temperatures, so we need to avoid cooling. The loss of power, or amount of cooling, is proportional to the atomic number. In other words, an element with more electrons will cause more cooling if sputtered, since there are more electrons being stripped.

6. Look up the low atomic number metals. Is there a material you think would be best for the irradiation challenge?

7. What might be a drawback of the material you chose? Does it have a high thermal conductivity?

1.5 Real Tokamaks

Now we have seen some challenges to building fusion reactors. Look into some of the current fusion projects. What materials do they use for their shield walls? Are they the same/similar to the ones you chose?

https://en.wikipedia.org/wiki/Tokamak#Experimental_tokamaks

<https://www.iter.org/machine/blanket>

<https://pmc.ncbi.nlm.nih.gov/articles/PMC6365859/>